

Krish Arora

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EDUCATION

University of British Columbia

Bachelor of Science in Mathematics (Science Co-op)

Vancouver, BC

Graduating May 2027

- Relevant Courses: Introduction to Artificial Intelligence, Machine Learning, Software Engineering, Data Structures & Algorithms, Object-Oriented Programming, Computer Systems, Relational Databases, Statistics

WORK EXPERIENCE

Software Developer Co-op

National Research Council Canada

January 2026 – April 2026

Saskatoon, SK

- Building a full-stack electronic lab notebook for NRC researchers with **21 CFR Part 11** regulatory compliance.
- Ensured data integrity across **24 DB models** by implementing immutable versioning and append-only audit trails.
- Reduced frontend bugs by managing all server state through **React Query hooks** with Zod schema validation.
- Hardening uploads by streaming files with **256-bit SHA** hashing and max-byte limits, ensuring attachment integrity.
- Secured **63+ endpoints** across **12 FastAPI routers** with service-layer RBAC and project-scoped audit enforcement.

PROJECTS

DreamCar | *Next.js, TypeScript, PostgreSQL, Prisma ORM, OpenAI API, Redis, TailwindCSS* [Repo](#) | [Demo](#)

- Built AI car recommendation engine with a deterministic 28-question quiz eliminating LLM latency from the core flow.
- Designed a Weighted scoring algorithm with **Prisma/PostgreSQL** and **Upstash Redis caching** for ranked results.
- Integrated **GPT-4o-mini** with structured JSON output for **natural-language** preference extraction on search page.
- Shipped the app on **Vercel** with shadcn/ui, Zod validation, TanStack Query, and **Vitest/Playwright** test coverage.

UBConnect | *React Native, TypeScript, Figma, Firebase, Jest*

[Repo](#) | [Demo](#)

- Engineered real-time social feed ranking algorithm mapping 41 interests to 11 categories for personalized event discovery.
- Modelled bidirectional social graph in Firestore with **5-state lifecycle**, enabling **O(1) mutual friend** lookups.
- Diagnosed and resolved **6 production bugs** across auth, avatar, and UI systems, deployed via zero-downtime EAS OTA.
- Enabled location-based event discovery by implementing **Firebase** authentication with **Google Maps API** integration.

POPGG | *React, Solidity, WebSocket, Figma*

[Repo](#)

- Visualized game analytics for 10,000+ players by building **React**-based web application with responsive scoring UI.
- Enabled democratic game improvements via **DAO** by building smart contract voting for player proposal submissions.
- Boosted developer-player trust by 45% measured by surveys by leveraging **blockchain** for transparent voting records.
- Increased user engagement by 30% by implementing real-time updates via **WebSocket** connection protocols.

PetAdopt BC | *Node.js, Express.js, MongoDB, Passport.js, Mocha, Chai*

[Repo](#) | [Demo](#)

- Built a full-stack pet adoption web platform with **40+ RESTful API endpoints** over **10 MongoDB collections**.
- Migrated legacy Oracle database to MongoDB Atlas, redesigning 10 schemas and preserving data integrity for 40+ APIs.
- Implemented JWT auth with role-based access control and Zod schema validation, securing all 40+ API write endpoints.
- Wrote **119 automated tests** via **Mocha, Chai & Supertest** with mongodb-memory-server, achieving 100% coverage.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL, Java, C++

Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Express, TailwindCSS, SASS, TanStack Query, Zod, shadcn/ui

Databases: PostgreSQL, MongoDB, Redis

Cloud & DevOps: Git, Docker, CI/CD Pipelines, Vercel, Linux

Testing: Jest, Vitest, Playwright, Pytest

AI/ML: GitHub Copilot, Claude Code, Deep Learning, Data Analysis

Other: Accessibility (a11y) practices - WCAG 2, Responsive design